

Routing benchmark: Solomon instances

T3 · Appendix A1–A2 · 56 instances · 100 customers per instance

Result: Omelet RouteOpt records a **+0.08% reported gap and 38/56 wins**. cuOpt records +1.76% and 13/56 wins, illustrating why the platform retains complementary solver backends.

Omelet technical report · measured engine performance. This report documents the experiments presented in the final-pitch appendix: measured results, solver comparisons, test conditions and interpretation. Public results are reported by Omelet; customer-level inputs and internal run records are not reproduced here. Benchmark execution is distinct from the NVIDIA integration roadmap.

Metric	Reported gap, instance win count and runtime.
Baseline / comparison	Omelet RouteOpt, cuOpt, RouteOpt-ALNS, LKH-3 and OR-Tools.
Setup	Public Solomon CVRPTW set; 56 × 100-customer instances; nominal 10-second budget. The appendix describes same-instance, same-machine comparisons; a separate Solomon machine/run configuration is not listed.
Gap reference	Best compared solver per instance, as defined in the full-results appendix; not an external best-known-solution database.
Date / reporting entity	Report: Omelet, 17 September 2026. The public appendix does not separately state the Solomon run date or individual experiment owner.

METHOD	HARDWARE	REPORTED GAP	WINS / 56	REPORTED TIME
Omelet RouteOpt	CPU	+0.08%	38	10.0 s
NVIDIA cuOpt	CPU + GPU	+1.76%	13	9.7 s
RouteOpt, ALNS configuration	CPU	+2.53%	3	8.1 s
LKH-3	CPU	+3.41%	2	47.6 s
OR-Tools	CPU	+4.56%	0	10.0 s

Time/feasibility exception: LKH-3 is reported feasible on 52/56 instances and at 47.6 seconds. This row is not an equal 10-second-budget result. Handling of infeasible cases in the aggregate gap is not specified.

Technical interpretation

Solver choice depends on the problem class and time budget: learned/hybrid RouteOpt leads this small-instance set, while the separate large-scale routing and MILP experiments show cuOpt strengths. Aggregate tables do not replace per-instance run records for reproduction.

[Open appendix A2: full results](#) · [Read the consolidated technical report](#)